$x' = \lambda x \qquad \lambda \in \mathbb{R}$

$\Downarrow$

$x(t) = C e^{\lambda t}, \quad C \in \mathbb{R}$

$t = x(t)$ fct. nec. $\qquad x' = x'(t)$ derivata fct. nec.

Sisteme de ecuații diferențiale liniare cu coeficienți constanți

$$\begin{cases} x_1' = a_{11} x_1 + \dots + a_{1n} x_n + b_1(t) \\ \dots \\ x_n' = a_{n1} x_1 + \dots + a_{nn} x_n + b_n(t) \end{cases}$$

t variabila

$x_1 = x_1(t), \dots, x_n = x_n(t)$ fct. nec.

$a_{11} \dots a_{nn} \in \mathbb{R}$

$b_1 \dots b_n : I \subseteq \mathbb{R} \to \mathbb{R}$ fct. cont.

$$A = (a_{ij})_{i,j=\overline{1,n}} \in \mathcal{M}_n(\mathbb{R})$$

$$X = X(t) = \begin{pmatrix} x_1(t) \\ \vdots \\ x_n(t) \end{pmatrix} : I \subseteq \mathbb{R} \to \mathbb{R}^n$$

$$b(t) = \begin{pmatrix} b_1(t) \\ \vdots \\ b_n(t) \end{pmatrix}$$

$b : I \to \mathbb{R}^n$

$$X' = AX + b(t)$$

$b(t) = 0$ — Sistem omogen

$b(t) \neq 0$ — Sistem neomogen

Sistem de ec. dif. liniare cu coef. constanți $X' = AX$

Thm. Problema Cauchy $\begin{cases} X' = AX \\ X(t_0) = X_0 \end{cases}$ are soluție unică $\forall t_0 \in \mathbb{R}$, $X_0 \in \mathbb{R}^n$.

Consec. Problema Cauchy $\begin{cases} X' = AX \\ X(t_0) = 0 \in \mathbb{R}^n \end{cases}$ are soluția unică $X(t) = 0$ $\forall t \in \mathbb{R}$

Prop. 1) $S_0 = \{ X : \mathbb{R} \to \mathbb{R}^n \mid X \text{ fct. dif.}, X' = AX \}$

spațiu vectorial peste $\mathbb{R}$ —

$X_1, X_2 \in S_0$

$X_1' = AX_1$, $X_2' = AX_2$ $\Big| \Rightarrow (X_1 + X_2)' = A(X_1 + X_2)$

$\Rightarrow X_1 + X_2 \in S_0$

(principiul superpoziției)

$\alpha \in \mathbb{R}$, $X \in S_0 \Rightarrow \alpha X \in S_0$

(exercițiu)

2) Fie $t_0 \in \mathbb{R}$ și $X_1, \dots X_p \in S_0$. Atunci:

$X_1 \dots X_p$ liniar indep $\Leftrightarrow$ $X_1(t_0), \dots, X_p(t_0) \in \mathbb{R}^n$ liniar indep.

3) $\dim_{\mathbb{R}} S_0 = n$

Consec. $\{X_1, \dots, X_n\}$ bază în S_0 $\Rightarrow$ soluția generală a sist. de ec. diferențiale $X' = AX$ este $\boxed{X(t) = C_1 X_1(t) + \dots + C_n X_n(t)}$ (sistem fundamental de soluții)

$C_1 \dots C_n \in \mathbb{R}$

$= \underbrace{(X_1(t) | \dots | X_n(t))}_{M(t) \text{ matrice } n \times n \text{ fundamentală de soluții (inversabilă)}} \begin{pmatrix} C_1 \\ \vdots \\ C_n \end{pmatrix}$

Problemă: cum găsim un sistem fundamental de soluții $X_1 \dots X_n$?

$n = 1$ $\quad x' = \lambda x \quad \Rightarrow \quad x(t) = Ce^{\lambda t}, \; C \in \mathbb{R}$

$n \geq 1$ $\quad X' = AX$ căutăm soluții de forma $X(t) = v e^{\lambda t}$, $v \in \mathbb{R}^n$

$X'(t) = v \cdot \lambda \cdot e^{\lambda t}$

$X' = AX \Rightarrow v \lambda e^{\lambda t} = A v e^{\lambda t} \Rightarrow \lambda v = Av$

$\Rightarrow \lambda \in \sigma(A)$

$v \in V_\lambda$

$X_1(t) = \underline{v_1 e^{\lambda_1 t}}, \dots, X_n(t) = \underline{v_n e^{\lambda_n t}}$ sistem fundamental de soluții

$\Leftrightarrow$ $\lambda_1, \dots \lambda_n \in \sigma(A)$ $\quad \Rightarrow$ $X_1(0) = v_1, \dots, X_n(0) = v_n \in \mathbb{R}^n$

$v_1 \in V_{\lambda_1}, \dots, v_n \in V_{\lambda_n}$ liniar independente

$\Leftrightarrow$

A diagonalizabilă.

$X' = AX$ sistem fundam. de sol. $X_i(t) = v_i e^{\lambda_i t}$, $i = \overline{1,n}$

$\Leftrightarrow$ A diagonalizabilă

$n = 2$ $\quad A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ $\quad X = \begin{pmatrix} x \\ y \end{pmatrix}$ $\quad \begin{cases} x' = ax + by \\ y' = cx + dy \end{cases}$

$\lambda_1, \lambda_2 \in \sigma(A)$ $\quad \lambda_1 \neq \lambda_2, \ \lambda_1, \lambda_2 \in \mathbb{R}$

$\lambda_1 = \lambda_2 \in \mathbb{R}$

$\lambda_1, \lambda_2 \in \mathbb{C} \quad \lambda_2 = \bar{\lambda}_1$

Caz 0 $\quad A = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$ $\quad \begin{cases} x' = \lambda_1 x \\ y' = \lambda_2 y \end{cases}$ (ec. sunt decuplate)

$\Rightarrow \begin{cases} x(t) = C_1 e^{\lambda_1 t} \\ y(t) = C_2 e^{\lambda_2 t} \end{cases}$ $\quad C_1, C_2 \in \mathbb{R}$

$X(t) = \begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = C_1 e^{\lambda_1 t} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + C_2 e^{\lambda_2 t} \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

exemplu $\quad \begin{cases} x' = x \\ y' = 2y \end{cases} \rightarrow \begin{cases} x(t) = C_1 e^t \\ y(t) = C_2 e^{2t} \end{cases}, \ C_1, C_2 \in \mathbb{R}$

$\lambda_1, \lambda_2 < 0$

$\lambda_1 < 0 < \lambda_2$

$0 < \lambda_1, \lambda_2$

$\lambda_1 = 0 < \lambda_2$

$\gamma : F(x, y) = 0 \quad \Leftrightarrow \quad \begin{cases} x = x(t) \\ y = y(t) \end{cases} \ t \in I$

$\int_\gamma \ldots$

Portret de faze

$C_1 = C_2 = 0 \Rightarrow x = y = 0$

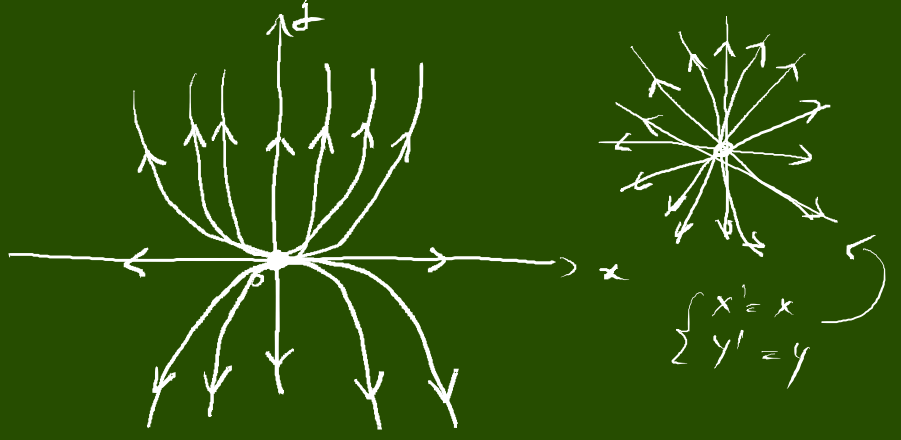
$C_1 = 0,\ C_2 \neq 0 \Rightarrow \boxed{x = 0}$

$y = y(t) = C_2 e^{2t}$

$C_1 \neq 0,\ C_2 = 0 \Rightarrow y = 0$

$x = x(t) = C_1 e^{t}$

$C_1, C_2 \neq 0 \Rightarrow \frac{x^2}{y} = \frac{C_1^2}{C_2} \Rightarrow y = \frac{C_2}{C_1^2} x^2$

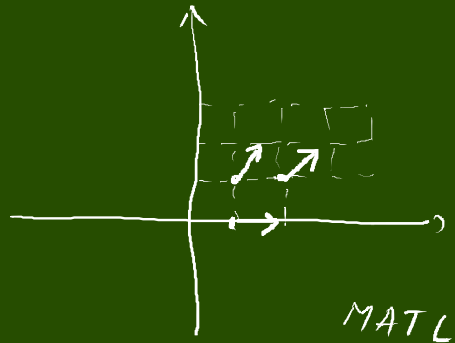


$$\begin{cases} x' = x \\ y' = 2y \end{cases}$$

$x = 1,\ y = 0 \Rightarrow x' = 1,\ y' = 0$

$x = 1,\ y = 1 \Rightarrow x' = 1,\ y' = 2$

$x = 2,\ y = 1 \Rightarrow x' = 2,\ y' = 2$



MATLAB (sem. II MN)

Cazul 1: A diagonalizabilă, $\lambda_1 \neq \lambda_2$, $\lambda_1, \lambda_2 \in \mathbb{R}$

$\dim_{\mathbb{R}} V_{\lambda_1} = \dim_{\mathbb{R}} V_{\lambda_2} = 1$

$V_{\lambda_1} = Sp\{v_1\}$, $V_{\lambda_2} = Sp\{v_2\}$

$\Rightarrow$ bază în spațiul soluțiilor S_0

$X_1(t) = e^{\lambda_1 t} v_1$, $X_2(t) = e^{\lambda_2 t} v_2$

$\Rightarrow$ soluția generală $X(t) = C_1 \underbrace{e^{\lambda_1 t} v_1}_{X_1(t)} + C_2 \underbrace{e^{\lambda_2 t} v_2}_{X_2(t)}$, $C_1, C_2 \in \mathbb{R}$

exemplu ecosistem – 2 specii în competiție pentru hrană

$x(t)$, $y(t)$ t - ani

$$\begin{cases} x' = 1,6x - 0,3y \\ y' = -0,8x + 1,4y \end{cases} \qquad x(0) = 10, \; y(0) = 10$$

$$A = \begin{pmatrix} 1,6 & -0,3 \\ -0,8 & 1,4 \end{pmatrix}$$

$$P_A(X) = X^2 - 3X + (2,24 - 0,24) = X^2 - 3X + 2$$

$1,6 \cdot 1,4 = 3,2.7$

$\lambda_1 = 1$ $\quad V_{\lambda_1} = \{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 \,/\, A\begin{pmatrix} x \\ y \end{pmatrix} = \lambda_1 \begin{pmatrix} x \\ y \end{pmatrix} \}$ $\quad \begin{cases} 1{,}6x - 0{,}3y = x \\ -0{,}8x + 1{,}4y = y \end{cases} \Rightarrow$

$\Rightarrow \quad -0{,}3y = -0{,}6x \Rightarrow y = 2x$

$V_{\lambda_1} = \{ \begin{pmatrix} x \\ 2x \end{pmatrix} / x \in \mathbb{R} \} = Sp\{ \underbrace{\begin{pmatrix} 1 \\ 2 \end{pmatrix}}_{v_1} \}$

$\lambda_2 = 2$ $\quad V_{\lambda_2} = \{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 \,/\, A\begin{pmatrix} x \\ y \end{pmatrix} = \lambda_2 \begin{pmatrix} x \\ y \end{pmatrix} \}$ $\quad \begin{cases} 1{,}6x - 0{,}3y = 2x \\ -0{,}8x + 1{,}4y = 2y \end{cases}$

$\Rightarrow \quad -0{,}3y = 0{,}4x \Rightarrow y = -4/3\,x$

$V_{\lambda_2} = \{ \begin{pmatrix} x \\ -4/3\,x \end{pmatrix} / x \in \mathbb{R} \} = Sp\{ \underbrace{\begin{pmatrix} 3 \\ -4 \end{pmatrix}}_{v_2} \}$

$\Rightarrow \quad X_1(t) = e^{\lambda_1 t} v_1 = e^t \begin{pmatrix} 1 \\ 2 \end{pmatrix}$, $X_2(t) = e^{\lambda_2 t} v_2 = e^{2t} \begin{pmatrix} 3 \\ -4 \end{pmatrix}$

$\Rightarrow \begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = X(t) = C_1 X_1(t) + C_2 X_2(t) = \underline{C_1 e^t \begin{pmatrix} 1 \\ 2 \end{pmatrix} + C_2 e^{2t} \begin{pmatrix} 3 \\ -4 \end{pmatrix}}$

$x(0) = 10 \quad y(0) = 10 \quad \Rightarrow \begin{cases} C_1 + 3C_2 = 10 & | \cdot 4 \\ 2C_1 - 4C_2 = 10 & | \cdot 3 \end{cases}$

$10C_1 = 70 \quad C_1 = 7 \quad , \quad C_2 = 1$

$$\Rightarrow \begin{cases} x(t) = 7ct + 3e^{2t} \\ y(t) = 14et - 4e^{2t} \end{cases} \xrightarrow{t \to \infty} \begin{matrix} \infty \\ -\infty \end{matrix}$$

$t_0 \quad y(t_0) = 0$

$14et = 4e^{2t}$

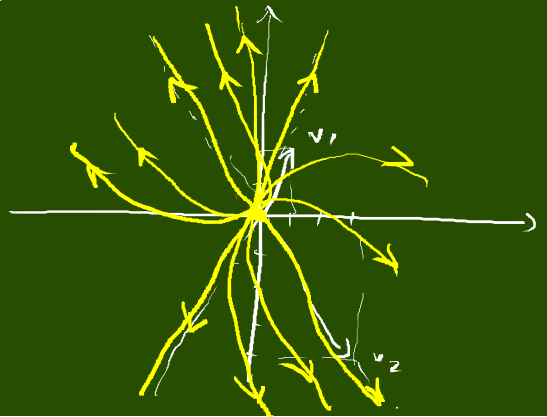
$\frac{7}{2} = et$

$1,\ldots \simeq \ln 3,5 = t$

$$\begin{cases} x(t) = C_1 e^t + 3C_2 e^{2t} \\ y(t) = 2C_1 e^t - 4C_2 e^{2t} \end{cases} \qquad C_1, C_2 \in \mathbb{R}$$

$$\begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = C_1 e^t \begin{pmatrix} 1 \\ 2 \end{pmatrix} + C_2 e^{2t} \begin{pmatrix} 3 \\ -4 \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 3 \\ 2 & -4 \end{pmatrix}}_{(v_1 | v_2)} \cdot \begin{pmatrix} C_1 e^t \\ C_2 e^{2t} \end{pmatrix}$$

$P_{B_c}^{B} = \{v_1, v_2\}$



Cas c 2 $\quad P_A(X) = (X-\lambda)^2$

$g(\lambda) = 2$

$g(\lambda) = 2 \Rightarrow A$ diagonalizable

$A = P \cdot \Delta \cdot P^{-1} = P \cdot \underbrace{\begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}}_{\lambda I_2} P^{-1} = \lambda I_2$

$\begin{cases} x' = \lambda x \\ y' = \lambda y \end{cases} \qquad x(t) = C_1 e^{\lambda t}, \; y(t) = C_2 e^{\lambda t}$

$g(\lambda) = 1$ $\quad A$ n c diagonalizable

Thm. Hamilton - Cayley

$P_A(A) = 0_2 \Rightarrow (A - \lambda I_2)^2 = 0_2$

$\dim_{\mathbb{R}} V_\lambda = 1 \Rightarrow V_\lambda \neq \mathbb{R}^2 \Rightarrow \exists\, v_2 \in \mathbb{R}^2$

$v_2 \notin V_\lambda$

Fie $v_1 = (A - \lambda I_2) v_2$

$\begin{cases} v_1 \neq 0 \quad (v_1 = 0 \Leftrightarrow (A - \lambda I_2) v_2 = 0 \\ \qquad\qquad \Leftrightarrow A v_2 = \lambda v_2 \Leftrightarrow v_2 \in V_\lambda \\ v_1 \in V_\lambda \quad (A v_1 = (A - \lambda I_2 + \lambda I_2) v_1 = \text{fals} \\ v_1, v_2 \text{ lin. indep.} \quad = (A - \lambda I_2) v_1 + \lambda v_1 \end{cases}$

$\left(\begin{matrix} v_1 \in V_\lambda \\ v_2 \notin V_\lambda \end{matrix} \right) \qquad = \underbrace{(A - \lambda I_2)^2 v_2}_{0_2} + \lambda v_1 = \lambda v_1 \Big)$

$\begin{cases} A v_1 = \lambda v_1 \\ A v_2 = (A - \lambda I_2) v_2 + \lambda v_2 = v_1 + \lambda v_2 \end{cases}$

$A \underbrace{(v_1 | v_2)}_{P \text{ inversabilă}} = (v_1 | v_2) \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}$

$A = P \cdot \underbrace{\begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}}_{\Delta} \cdot P^{-1}$

Jordan

$X' = AX$

$X_1(t) = e^{\lambda t} v_1 \in S_0$

$X_2(t) = e^{\lambda t}(tv_1 + v_2)$

$X_2 \in S_0$

X_1, X_2 lin. indep ($\Rightarrow$ base in S_0)

$X_2'(t) = \lambda e^{\lambda t}(tv_1 + v_2) + e^{\lambda t} v_1$

$\lambda e^{\lambda t}(tv_1 + v_2) + e^{\lambda t} v_1 = A e^{\lambda t}(tv_1 + v_2)$

$\lambda t v_1 + \lambda v_2 + v_1 = A t v_1 + A v_2$

$\lambda t v_1 + \lambda v_2 + v_1 = t\lambda v_1 + v_1 + \lambda v_2$

$\Rightarrow X_2 \in S_0$

X_1, X_2 lin. indep. $\iff$ $\underbrace{X_1(0)}_{v_1}, \underbrace{X_2(0)}_{v_2}$ lin. indep. ✓

$\Rightarrow \quad X_1(t) = e^{\lambda t} v_1, \quad X_2(t) = e^{\lambda t}(t v_1 + v_2)$ S.f. fundamental de soluções p/ $X' = AX$

$\Rightarrow$ sol. geral $\quad X(t) = \begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = C_1 e^{\lambda t} v_1 + C_2 e^{\lambda t}(t v_1 + v_2)$

$C_1, C_2 \in \mathbb{R}$.

exemplo: $\quad A = \begin{pmatrix} 2 & -1 \\ 1 & 4 \end{pmatrix} \qquad \lambda = 3$

$\operatorname{tr} A = 6$

$\det A = 9$

$$\begin{cases} x' = 2x - y \\ y' = x + 4y \end{cases} \quad \begin{cases} x(0) = 0 \\ y(0) = 0 \end{cases} \Rightarrow \begin{matrix} x(t) = 0 \\ y(t) = 0 \end{matrix} \quad \forall t$$

$$\begin{cases} x(0) = 1 \\ y(0) = 0 \end{cases}$$

$\lambda = 3 \qquad V_\lambda : \quad \begin{cases} 2x - y = 3x \\ x + 4y = 3y \end{cases} \Rightarrow \underline{x = -y} \qquad V_\lambda = \mathrm{Sp}\left\{\begin{pmatrix} 1 \\ -1 \end{pmatrix}\right\}$

$\left\{ \right. \quad v_2 \in \mathbb{R}^2 \quad v_2 \notin V_\lambda \qquad \underline{v_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}} \qquad \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$$v_1 = (A - 3I_2)v_2 = \left(\begin{pmatrix} 2 & -1 \\ 1 & 4 \end{pmatrix} - \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}\right)\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 & -1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$= \begin{pmatrix} -2 \\ 2 \end{pmatrix} \in V_1$$

$$\Rightarrow \begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = c_1 e^{3t}\begin{pmatrix} -2 \\ 2 \end{pmatrix} + c_2 e^{3t}\left(t\begin{pmatrix} -2 \\ 2 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix}\right)$$

$$\begin{aligned} x(0) &= 1 \\ y(0) &= 0 \end{aligned} \quad \Rightarrow \quad \begin{aligned} -2c_1 + c_2 &= 1 \\ 2c_1 + c_2 &= 0 \end{aligned}$$

$$c_2 = 1/2 \qquad c_1 = -1/4$$

$$\Rightarrow \begin{cases} x(t) = 1/2\, e^{3t} + 1/2\, e^{3t}(-2t + 1) \\ y(t) = -1/2\, e^{3t} + 1/2\, e^{3t}(2t + 1) \end{cases}$$